LITERATURE REVIEW & PROJECT EVALUATION

Superstore Sales Analytics & Dashboard Architecture

Project Context & Dataset Overview

This Literature Review evaluates the Superstore Sales Analysis project based on the transaction-level dataset supplied. The dataset provides continuous sales coverage across four key operational dimensions: temporal, geographic, customer segment, and product hierarchies.

Dataset Metric	Observed Value	Analytical Scope & Significance
Records & Attributes	9,800 records | 18 columns	Provides high granularity for multidimensional sales slicing and trend discovery.
Transactions & Entities	4,922 unique orders | 793 unique customers	Enables order-level frequency metrics and customer purchasing behavior analysis.
Temporal Coverage	03 Jan 2015 – 30 Dec 2018 (4 Years)	Supports robust Year-over-Year (YoY) and multi-year seasonal pattern modeling.
Financial Metrics	$2,261,536.78 Total Sales	Establishes baseline sales performance across categories and regional hubs.

Dataset Metric	Observed Value	Analytical Scope & Significance
Data Quality & Hygiene	11 missing Postal Codes; 0 missing values elsewhere	Data cleanliness is exceptionally high, requiring minimal imputation during ETL.

1. Feedback & Evaluation

Lecturer Assessment Framework: This evidence-based evaluation assesses the execution quality, analytical depth, and data-driven insights derived from the Superstore dataset.

Evaluation Area	Evidence-Based Findings & Metrics	Lecturer Assessment & Feedback
Dataset Structure & Ingestion	Successfully ingested 9,800 rows across 18 fields. Isolates transactions across 4 regions, 3 customer segments, and 3 product categories.	Demonstrates a strong foundational grasp of data ingestion and attribute mapping. The dataset is well-structured for exploratory analytics.
Product Performance Analysis	**Technology** dominates overall revenue at **$827,455.87 (36.6%)**, followed by Furniture ($728,658.58) and Office Supplies ($705,422.33).	Excellent categorical aggregation. Insights clearly identify revenue drivers and product portfolio dependencies.
Customer Segmentation Insights	The **Consumer segment** generates the largest share at	Clear identification of market share distribution. Highlights

Evaluation Area	Evidence-Based Findings & Metrics	Lecturer Assessment & Feedback
	$1,148,060.53, outperforming Corporate ($688,494.07) and Home Office ($424,982.18).	core target audiences effectively.
Regional Revenue Variance	The **West region** leads performance with **$710,219.68**, whereas the **South region** underperforms at **$389,151.46**.	Strong regional comparative breakdown. Establishes a concrete baseline for geographical expansion or resource allocation strategies.
Temporal Trend Analysis	Sales demonstrate consistent growth, expanding from **$459,436.01 in 2016** to **$722,052.02 in 2018**.	Effective longitudinal analysis. Trends are clearly validated across yearly intervals without making unverified causal claims.
Data Quality & Scope Boundaries	Handled 11 missing Postal Codes. The dataset lacks explicit Cost, Profit, Unit Quantity, and Discount fields.	Commendable transparency regarding data limitations. Appropriately avoids treating top-line revenue as bottom-line profit.

2. Suggested Improvements

To elevate the project to enterprise-grade analytics standards, the following technical and architectural enhancements are recommended:

- **Data Preparation & ETL Documentation:** Formally document all data transformation steps in Power Query (e.g., date parsing, data type declarations,

duplicate verifications, and Postal Code handling) to ensure full pipeline reproducibility.

- **Standardized KPI Architecture:** Formulate explicitly defined Key Performance Indicators (KPIs) including Total Revenue, Average Sales per Record ($230.77), Unique Order Volume (4,922), Active Customer Count (793), and Year-over-Year (YoY) Growth percentage.
- **Data Modeling & Schema Optimization:** Transition from flat-file aggregation to a dimensional Star Schema in Power BI. Construct separate Dimension tables (`Dim_Customer`, `Dim_Product`, `Dim_Geography`, and a dedicated `Dim_Date`) linked cleanly to a central `Fact_Sales` table.
- **Advanced Time Intelligence DAX:** Implement DAX measures for Year-to-Date (YTD), Same Period Last Year (SPLY), dynamic ranking, and rolling 12-month moving averages to provide actionable time-series insights.
- **Interactive User Experience (UX):** Incorporate responsive visual slicers for Order Date, Region, Category, Customer Segment, and Ship Mode. Ensure seamless cross-filtering across all visual cards and charts.
- **Analytical Depth & Pareto Modeling:** Introduce Pareto (80/20) analysis to identify top-performing products and customer tiers contributing to the majority of total revenue.
- **Handling Data Gaps:** Formally request or integrate supplementary Cost and Profit datasets to calculate true Profit Margins and Discount Impacts, ensuring financial performance isn't overstated.
- **Validation & Automated Testing:** Implement testing protocols to verify that visual dashboard aggregates match raw source totals across all applied filter contexts.

3. Final Grading Criteria

The evaluation framework is structured across four core operational pillars, focusing on qualitative excellence, analytical rigor, and technical execution without static numerical point allocations:

Criteria Pillar	Qualitative Evaluation Scope & Assessment Focus
Documentation & Literature Review	Evaluates the clarity, thoroughness, and professional structure of project documentation uploaded to GitHub. Focuses on dataset context, metric definitions, architectural clarity, data hygiene logs, and explicit acknowledgment of dataset limitations.
Implementation & Data Modeling	Assesses technical execution across Power Query, Excel, and Power BI. Evaluates code cleanliness, schema optimization (Star Schema), precision of DAX measures, data type accuracy, dynamic slicer functionality, and visual dashboard usability.
Testing & Quality Assurance	Measures the rigor of data validation protocols. Evaluates edge-case testing, filter cross-verification, reconciliation of visual KPI cards against raw CSV source data, and error-handling routines.
Presentation & Insights Communication	Appraises the executive quality of visual storytelling, logical layout flow, presentation aesthetics, ability to translate complex numerical metrics into actionable business recommendations, and technical responsiveness during project defense.